FIELDS, COMPLETE LOCAL RINGS, THE INTEGERS, ONE-DIMENSIONAL LOCAL AND NAGATA RINGS, AND DEDEKIND DOMAINS OF CHARACTERISTIC ZERO ARE J-2

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Allowed standard prerequisites (editorial).

- The two criteria for a ring to be J-2: the condition on finite purely inseparable extensions of residue fields, and the condition on finite ring extensions being J-1. Lemma \ref{lemma-J-2}, cited at proof lines 11959-11960, 11978-11979, and 11987-11988. Provide the criterion that is verified for each of the seven classes.
- The integral closure of a one-dimensional Nagata domain in a finite extension is a normal domain of dimension one, hence regular. Algebra, Lemma \ref{algebra-lemma-integral-dim-up} and Algebra, Lemma \ref{algebra-lemma-criterion-normal}, cited at proof lines 11968-11971. Settle the case of a minimal prime in a one-dimensional Nagata ring.
- A finite extension of a complete local ring is a product of complete local rings. Algebra, Lemma \ref{algebra-lemma-quotient-complete-local}, cited at proof lines 11980-11981. Reduces the complete local case to the case of a domain via the regular-locus criterion.
- A Noetherian complete local domain is J-0, and regularity is preserved when the regular locus meets every closed subset in a nonempty open subset. Lemma \ref{lemma-complete-Noetherian-domain-J-0} and Lemma \ref{lemma-intersection-regular-with-closed}, cited at proof lines 11982-11984. Finish the complete local case.

Source reference. Original lemma 07PJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9799–9799. The lemma's own canonical tag, quoted

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so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07PC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9792–9792. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00OJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 897–897. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 031S: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3995–3995. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0325: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4008–4008. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07P9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9789–9789. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07PI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9798–9798. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Proposition 0.2. *The following types of rings are J-2:*

(1) *fields,*
(2) *Noetherian complete local rings,*
(3) $\mathbf{Z}$*,*
(4) *Noetherian local rings of dimension* 1*,*
(5) *Nagata rings of dimension* 1*,*
(6) *Dedekind domains with fraction field of characteristic zero,*
(7) *finite type ring extensions of any of the above.*

Proof. For cases (1), (3), (5), and (6) this is proved by checking condition (4) of Lemma 07PC. We will only do this in case R is a Nagata ring of dimension 1. Let $\mathfrak{p} \subset R$ be a prime ideal and let $L/\kappa(\mathfrak{p})$ be a finite purely inseparable extension. If $\mathfrak{p} \subset R$ is a maximal ideal, then $R \to L$ is finite and L is a regular ring and we've checked the condition. If $\mathfrak{p} \subset R$ is a minimal prime, then the Nagata condition insures that the integral closure $R' \subset L$ of R in L is finite over R. Then R' is a normal domain of dimension 1 (Algebra, Lemma 00OJ) hence regular (Algebra, Lemma 031S) and we've checked the condition in this case as well.

For case (2), we will use condition (3) of Lemma 07PC. Let R be a Noetherian complete local ring. Note that if $R \to R'$ is finite, then R' is a product of Noetherian complete local rings, see Algebra, Lemma 0325. Thus by Lemma 07P9 it suffices to prove that a Noetherian complete local ring which is a domain is J-0, which is Lemma 07PI.

For case (4), we also use condition (3) of Lemma 07PC. Namely, if R is a local Noetherian ring of dimension 1 and $R \to R'$ is finite, then $\mathrm{Spec}(R')$ is finite. Since the regular locus is stable under generalization, we see that R' is J-1. $\square$